

Pennada Venkata Chaitanya

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[Portfolio](#)

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WORK EXPERIENCE

• Associate Software Engineer | Innocito

April 2025 - Present

- Built a real-time **AI Voice Agent** integrating LLMs, speech-to-text, and text-to-speech, reducing response latency by **3 seconds per request** to enable more natural conversations.
- Built **zero-touch, production-ready data pipelines** using Prefect, FastAPI, and MongoDB, leveraging **async and concurrent processing** to automate large-scale data extraction and ingestion workflows and achieve up to a **12× performance improvement**.
- Researched and developed **MCP servers** for local and remote environments, implementing emerging AI infrastructure with limited documentation.

• AI Software Developer | Freelance

March 2024 - February 2025

1 Year

- For **Hooman Digital**, fine-tuned AI agents for AI-powered social media, cutting manual task time by **30%** via automation.
- Built an automated AI GIF Studio, achieving **95% accuracy**, **5× speed**, and **100× engagement**.
- For **AI Gen Media**, developed an AI-driven text-to-video workflow that achieved **80% visual consistency** across generated scenes.

• SDE AI/ML Intern | Spring Works

July 2023 - Sept 2023

3 Months

- Developed and delivered chatbot proof-of-concepts while improving automated **test coverage from 30% to 75%** using Pytest and collaborating with the engineering team.

EDUCATION

• B.Tech in Computer Science

2020 - 2024

- Graduated from Aditya College of Engineering and Technology with a focus on AI & ML.

VOLUNTEER EXPERIENCE

• SDE Mentor | Innovative Origin

April 2024 - March 2025

1 Year

- Mentored **100+ students** in AI & ML, focusing on practical projects & industry applications.

SKILLS & PROFICIENCIES

- **Programming Languages:** Python, SQL, C, C++
- **Generative AI & Agent Systems:** Generative AI, AI Agents, RAG, Multimodal LLMs, LangGraph, LangChain, LlamaIndex, MCP, Vector Databases, Ollama
- **Machine Learning & Vision:** PyTorch, NLP, Computer Vision, OpenCV
- **Backend & Data:** FastAPI, PostgreSQL, MongoDB, SQLite, Prefect, REST APIs
- **Engineering Tools:** asyncio, Playwright, Pytest, Git, GitHub, Streamlit

PROJECTS

• Synapse

[Website](#) | [GitHub](#)

- Architected and built an **offline-first AI knowledge system** with block-based notes, tasks management, semantic retrieval, & AI-driven actions.
- Designed **local-cloud sync** with SQLite & PostgreSQL, using block-level incremental updates & HLC-based conflict resolution.
- Built **hybrid local-cloud RAG & AI pipelines** with incremental vector indexing and tool execution for contextual retrieval & system actions.
- **Tech Stack:** Python, FastAPI, PostgreSQL, SQLite, LangGraph, RAG, Vector Databases, Ollama, MCP.

• Ed AI

[GitHub](#)

- An AI assistant offering course recommendations, Socratic AI chat, and mock interview simulations to address learning and teaching challenges.
- Achieved **2x faster execution** by optimizing & fine-tuning AI algorithms and boosting efficiency.
- **Tech Stack:** Python, FastAPI, LangChain, RAG, Streamlit, Generative AI, Pytest.

ACHIEVEMENTS

- Won Hackathon: '**Solving for India**' by GFG, Google and AMD at institutional region. [Certificate](#)
- Won [devpost's](#) International Hackathon 'Fresh Hacks'.
- Earned **5-star** Problem Solving rating on [HackerRank](#) and solved **300+ problems** across [Leetcode](#), [GFG](#).

CERTIFICATIONS

- Azure AI Fundamentals [Certification](#)
- MTA Python [Certification](#)